

CSCE 581: Introduction to Trusted AI

Week 15 - Lectures 29: Final Session

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

30 APRIL, 2026

Carolinian Creed: “I will practice personal and academic integrity.”

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Organization of Lectures 30

- Main Message of Course
- Finals - Q3
- Concluding Section
 - Ask me anything

17	Mar 17 (Tu)	Invited Guest –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2 - end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12] Project – Sprint 2 - end
22	Apr 2 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods	Project – Sprint 3 - start
23	Apr 7 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods	[Week 13] Quiz 3 - end
24	Apr 9 (Th)	Explanation Methods Trust: AI Testing	
25	Apr 14 (Tu)	Trust: Human-AI Collaboration	[Week 14] Quiz 3 - end
26	Apr 16 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World	
27	Apr 21 (Tu)	Project presentation	[Week 15] Project – Sprint 3 - end
28	Apr 23 (Th)	Project presentation	Last day of class
	Apr 28 (Tu)		Reading Day
29	April 30 (Th)	12:30pm – Final exam/ Overview	[Week 16] Optional, information shared

#1 NEWS — Talkies (LLM trained on 1930 era data)

Link - <https://www.marktechpost.com/2026/04/27/meet-talkie-1930-a-13b-open-weight-llm-trained-on-pre-1931-english-text-for-historical-reasoning-and-generalization-research/>

- **Context**
 - 13-billion parameter open-weight language model trained exclusively on pre-1931 English text
 - **December 31, 1930** — date when works enter the public domain in the United States, making pre-1931 text legally usable for training.
- **Findings**
 - Events after 1930 — talkie’s knowledge cutoff — are consistently more surprising to the model, with the effect most pronounced for 1950s and 1960s events, followed by a plateau.
 - This creates a principled setup for studying how forecasting ability scales with model size and how performance decays over longer temporal horizons.
 - Allows clean experimental setting to test how well an LM can generalize beyond its pre-training data. For example, the team tested whether talkie could learn Python — a language that didn’t exist in 1930 — by providing a few in-context demonstration examples. Using the HumanEval benchmark, they found that while vintage models dramatically underperform web-trained models, they are “slowly but steadily improving at this task with scale.”
- **Insight**
 - New possibilities for research if we have data and know how to use.
 - Another example: Lighthouse (Civil Rights Explorer at USC)

High Level Plan (Original)

CSCE 581 –

- Week 1: Introduction
- Week 2: Background: AI - Common Methods
- Week 3: The Trust Problem
- **Week 4: Machine Learning (Structured data) - Classification**
- **Week 5: Machine Learning (Structured data) - Classification – Trust Issues**
- **Week 6: Machine Learning (Structured data) – Classification – Mitigation Methods**
- **Week 7: Machine Learning (Structured data) – Classification – Explanation Methods**
- Week 8: Machine Learning (Text data) – Classification, **Large Language Models**
- **Week 9: Machine Learning (Text data) - Classification – Trust Issues**
- **Week 10: Machine Learning (Text data) – Classification – Mitigation Methods**
- **Week 11: Machine Learning (Text data) – Classification – Explanation Methods**
- Week 12: Emerging Standards and Laws
- Week 13: Project presentations
- Week 14: Project presentations, Conclusion

AI/ ML topics and with a
focus on fairness, explanation,
Data privacy, reliability

Student Assessment

A = [920-1000]
 B+ = [870-919]
 B = [820-869]
 C+ = [770-819]
 C = [720-769]
 D+ = [670-719]
 D = [600-669]
 F = [0-599]

Tests	Undergrad	Grad
Course Project – report, in-class presentation	600	600
Quiz – best of 2 from 3	200	200
Final Exam	200	100
Additional Final Exam – Paper summary, in-class presentation		100
Total	1000 points	1000 points

Grades and Attendance Policy

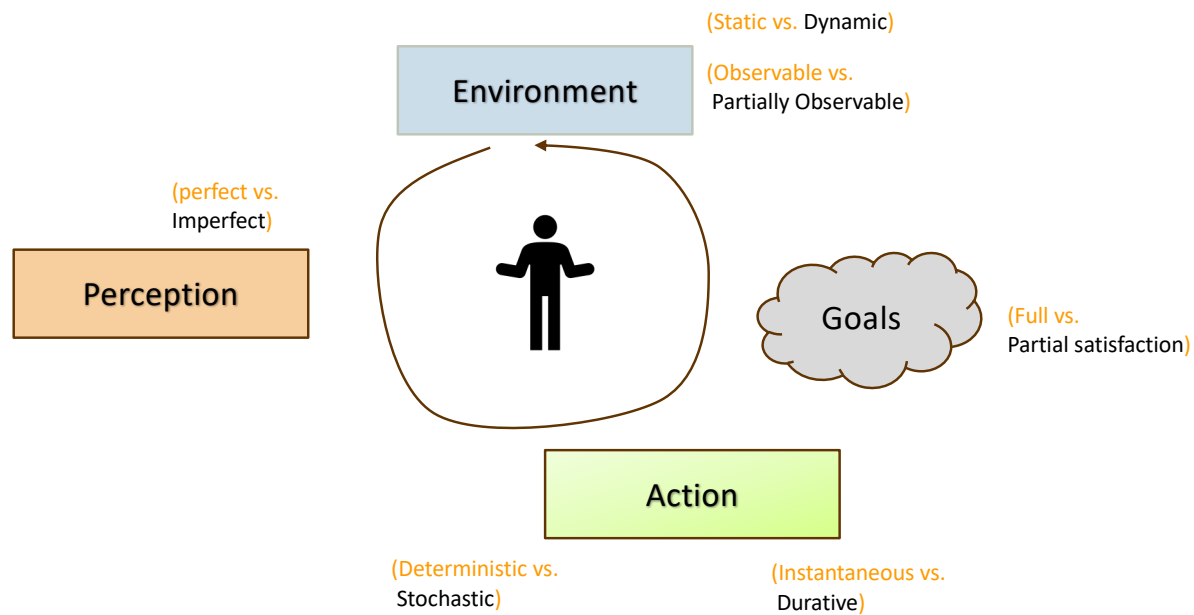
As an in-person class, I will impose a common-sense, **necessary – but not sufficient**, attendance policy for those taking the class for grades.

will be out of 14 classes:

1. 80% or more classes attending: **A, B+ or B**
2. 50-79% classes: **B+, B, C+, C, D+ or D**
3. 49% or below: **F**

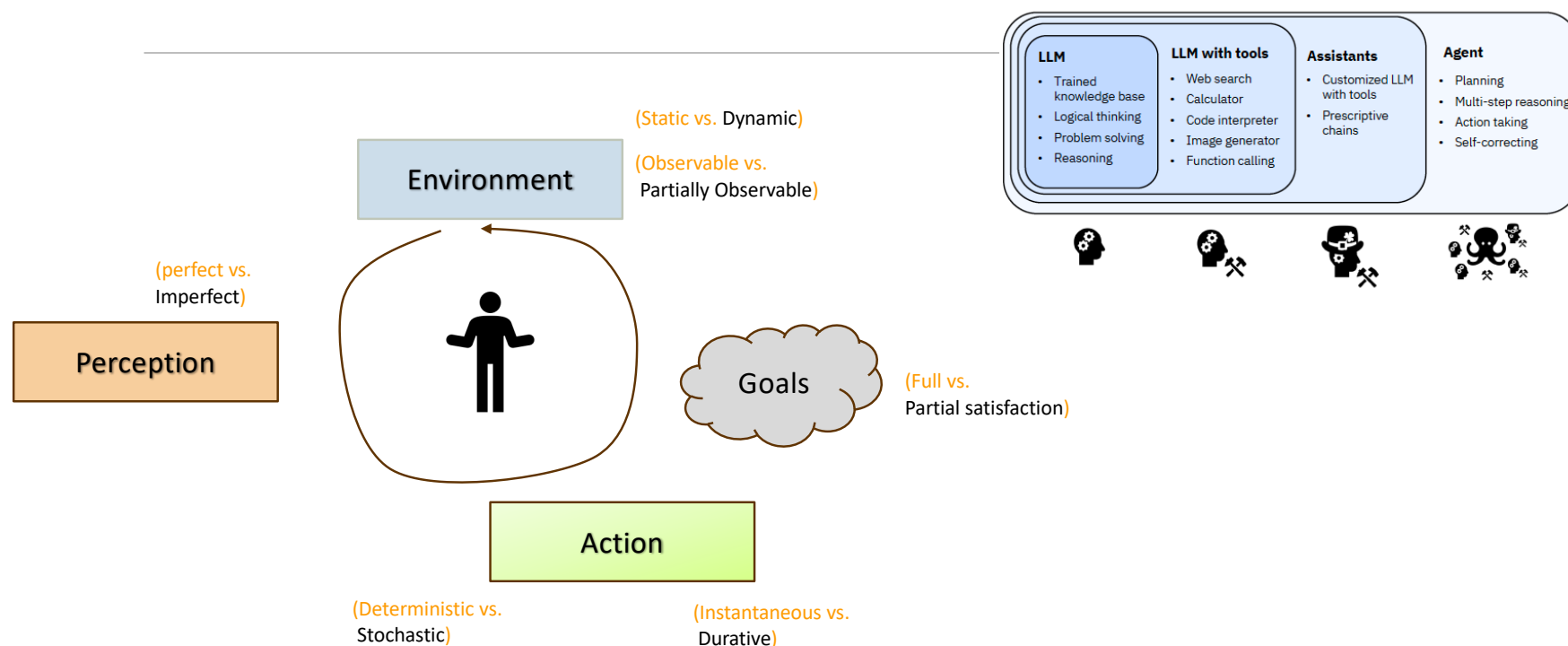
Main Message From Course

Intelligent Agent Model

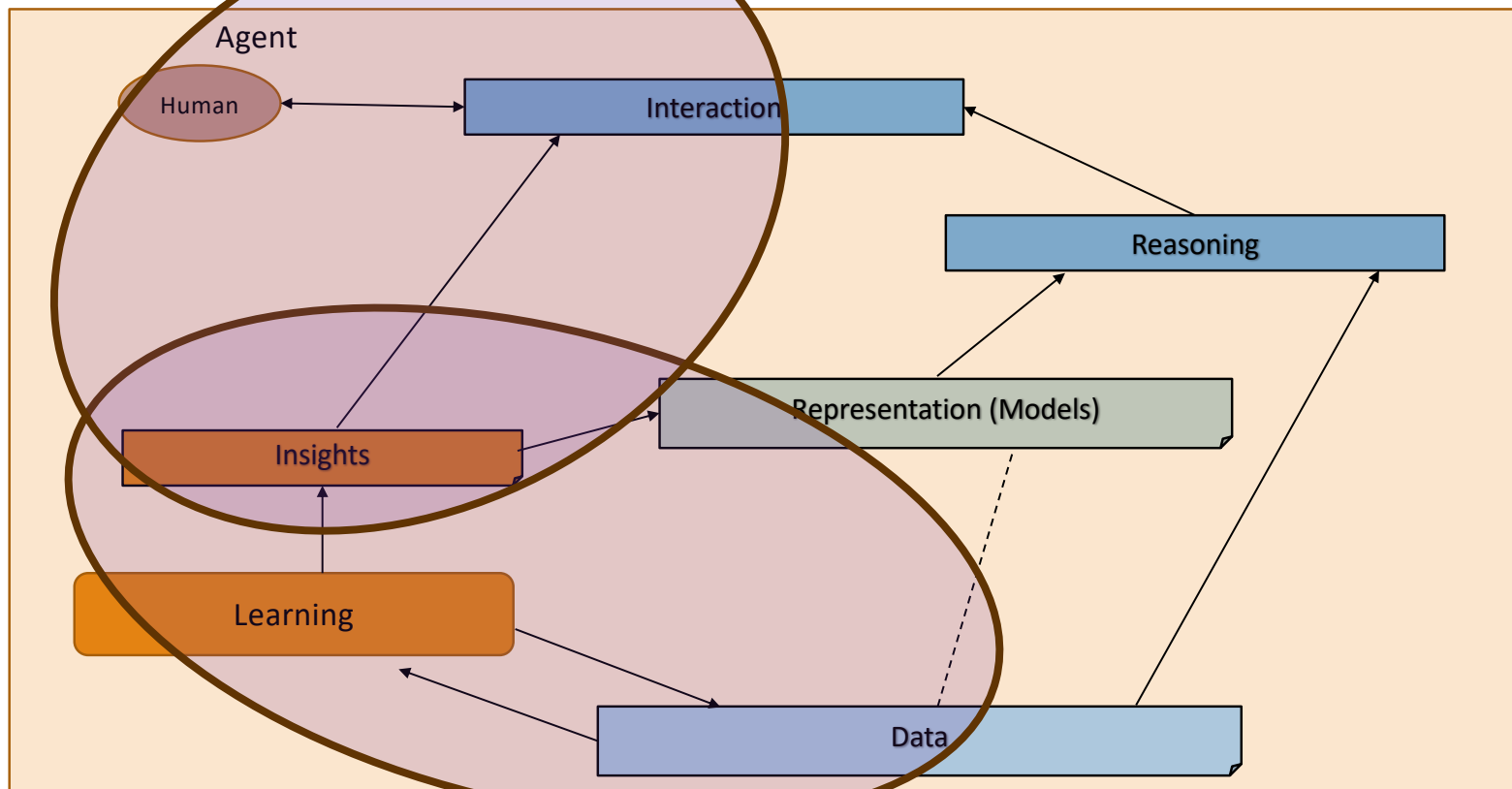


Intelligent Agent Model vis-a-vis Agentic AI

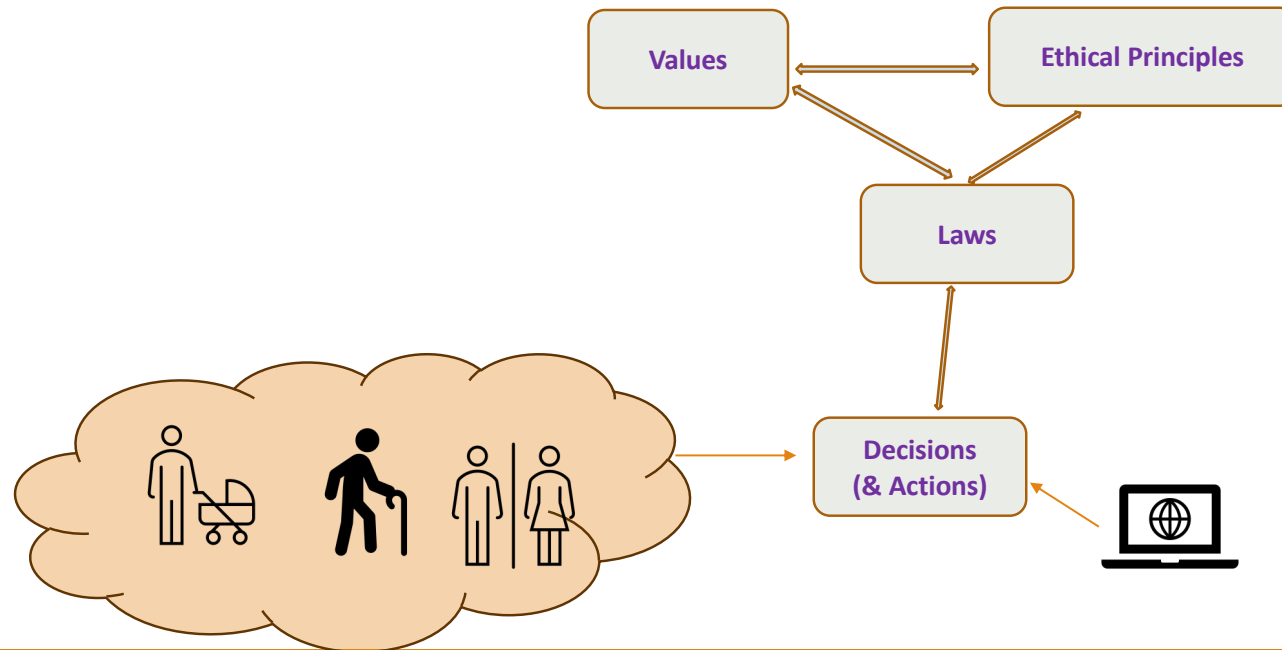
Credit: K. Varshney's talk - [Individual and Collective Human Agency in the Face of 'AI'](#), 2026



Relationship Between Main AI Topics (Covered in Course)



Values, Ethics, Law, Decisions (Actions) with Trust



Trust: how can, humans or machines (AI) or in collaboration together, take decisions and actions following the same **values**, **ethical** principles and **laws** ?

Assumption: they should follow same values, principles and laws.

Main AI Ethics Issues



DATA GOVERNANCE
AND PRIVACY



FAIRNESS AND
INCLUSION



HUMAN AND
MORAL AGENCY



VALUE ALIGNMENT



ACCOUNTABILITY



TRANSPARENCY AND
EXPLAINABILITY



TECHNOLOGY
MISUSE

Credits:

Tutorial on [Trusting AI by Testing and Rating Third Party Offerings](#) at IJCAI 2020, Biplav Srivastava, Francesca Rossi, Jan 2021

Finals, Q3

Case to Consider

Choice 1

- Role: Chief AI Officer @University
- Candidates: Any faculty who has AI in their biodata and whose name comes up in University's (e.g., sc.edu) web page

Choice 2

- Role: Coach for University Team in a college sports (e.g., basketball)
- Candidates: Any coach in university team across the US

Ask-Me-Anything (AMA)
